

PRINCIPLE OF CONTINUOUS CASTING

Molten bronze is poured from a furnace into a crucible which is cemented to the neck of a graphite die. The lower half of the die is water cooled, the metal sets in this section and is withdrawn through two sets of rollers. A flying saw mounted below the roller cuts the bar to the required length. Continuous cast rod has greatly improved physical properties compared with normal sand-cast material. Because of the bottom pouring conditions and rapid cooling the metal receives in the cooling zone, inclusions are completely eliminated, grain structure is tight and even giving excellent pressure tightness qualities. Tensile strength and wearing qualities are increased by up to 35%.

SPECIFICATIONS

Our standard stock bronze is listed below. Other specifications can be supplied on request provided sufficient quantity is ordered.

Material Specification:

British Standard	-----BS.1400 LG2-C
Japanese Standard	-----JIS H5111 BC6C
American Standard	-----ASTM SAE 40
	ASTM B145-52.4A
	ASTM 584-85a C83600

Approximate Chemical Composition

-----Copper 85%

Tin 5%

Zinc 5%

Lead 5%

Typical Mechanical Properties : U.T.S.

-----18 Tons/in²

28 kg/mm²

17 Tons/in² BS.1400(min)

Yield Stress

-----10 Tons/in²

15 kg/mm²

Elongation

-----25%

15% BS.1400(min)

Hardness (Brinell No.)

-----80

ADVANTAGES

Constant research and improved production techniques ensure top quality material with many advantages :-

- ☐ Reduced machining allowance - less waste
- ☐ Reduced tool wear
- ☐ Higher machining speeds
- ☐ Excellent mechanical properties
- ☐ Freedom from gas and porosity
- ☐ Close grain structure provides good pressure resistance

APPLICATION

General purpose bearings, Automotive and Machine

- Hydraulic Components
- Gears, Worm Wheels
- Valve seats
- Slide valves
- Rock drill parts
- Crusher bearings
- Hose nozzles
- Irrigation equipment
- Rollers
- Washers
- Wear rings